BBMS GHANA

Cloud-Based Blood Bank Management System

Blood Bank Management Report

Comprehensive Inventory & Activity Analysis

Facility: All Registered Facilities
Reporting Period: All recorded data
Generated: 08 July 2026 at 11:11

Executive Summary

This report presents a comprehensive analysis of blood bank operations at **All Registered Facilities** for the period **All recorded data**. The report covers inventory status, blood group distribution, component breakdown, internal and network request activity, storage utilization, and a recent activity audit trail. The data is intended to support operational decision-making, regulatory compliance, and continuous improvement of transfusion services.

Key Metrics at a Glance

TOTAL UNITS	AVAILABLE	RESERVED	ISSUED
190	**127**	**5**	**7**

As of this report, the facility holds a total of **190** blood units across all blood groups and component types. Of these, **127** units (66.8%) are currently available for transfusion, while **52** units have expired and **1** have been discarded. A further **14** units are expiring within the next 5 days and require immediate prioritization for use. The presence of 52 expired units indicates that expiry monitoring protocols should be reviewed to minimize future wastage. These figures form the baseline for the detailed analysis presented in the following sections.

1. Blood Group Stock Distribution

The chart below illustrates the current stock levels for each of the eight major blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-). Each group is broken down by status: **Available** (ready for transfusion), **Reserved** (held for a pending request), and **Expired** (no longer safe for use). This visualization enables blood bank officers to quickly identify which blood groups are well-stocked and which may require urgent replenishment or a network broadcast request.

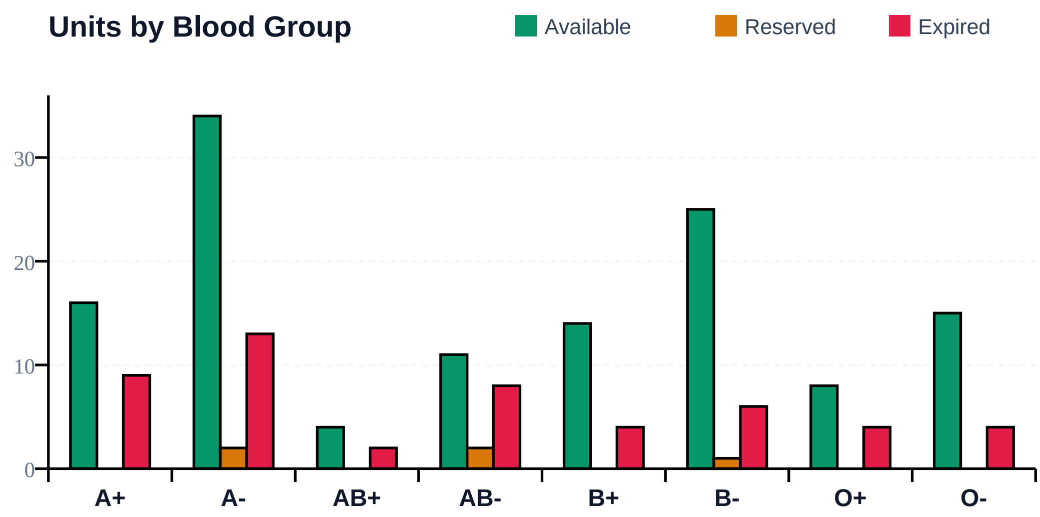


Figure 1: Blood group stock distribution by status

Detailed Breakdown by Blood Group

The table below provides exact figures for each blood group, including total units registered, units currently available, expired units, reserved units, and units already issued for transfusion. The final row presents the facility-wide totals.

Blood Group	Available	Total	Expired	Reserved	Issued
A+	16	25	9	0	0
A-	34	53	13	2	3
AB+	4	6	2	0	0
AB-	11	21	8	2	0
B+	14	19	4	0	1
B-	25	34	6	1	2
O+	8	13	4	0	1
O-	15	19	4	0	0
TOTAL	**127**	**190**	**50**	**5**	**7**

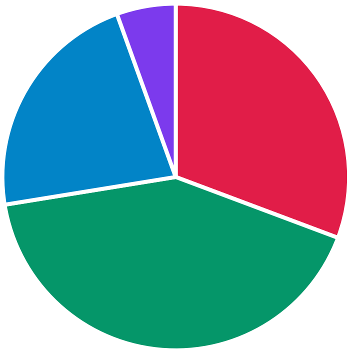
Table 1: Detailed blood group stock breakdown

Interpretation: The blood group with the highest available stock is **A-** with 34 units. The following blood groups are below the minimum stock threshold of 5 units and may require attention: **AB+**. Consider prioritizing these groups for the next donor recruitment drive or initiating a network broadcast request if clinical demand is anticipated.

2. Component Type Distribution

Blood is processed into four primary component types, each with specific clinical uses and shelf lives: **Whole Blood** (35 days), **Red Blood Cells** (42 days), **Platelets** (5 days), and **Fresh Frozen Plasma** (1 year). The pie chart below shows the proportional distribution of currently available units across these component types, helping staff ensure a balanced inventory that meets diverse clinical needs.

Available Units by Component Type



Red Blood Cells	39 units (31%)
Fresh Frozen Plasma	53 units (42%)
Whole Blood	28 units (22%)
Platelets	7 units (6%)

Figure 2: Available units by component type

Interpretation: Fresh Frozen Plasma represents the largest portion of available inventory at 53 units (42% of available stock). Platelets represent a small fraction of inventory, which is expected given their short 5-day shelf life. Close monitoring is recommended to avoid shortages.

3. Internal Blood Requests

Internal blood requests are submitted by nurses and doctors from clinical wards when a patient requires a transfusion. Each request specifies the blood group, quantity, urgency level, and patient reference. The Blood Bank Officer reviews each request, identifies a compatible available unit, and either approves, issues, or rejects the request. The chart below shows the current distribution of internal requests by their processing status.

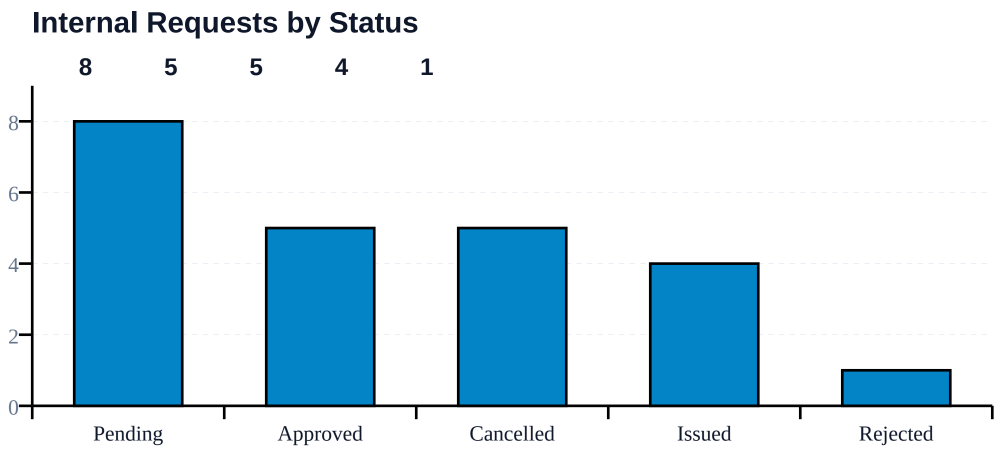


Figure 3: Internal blood requests by processing status

Interpretation: A total of **23** internal requests were recorded during the reporting period. Of these, **4** were fulfilled (issuance rate: 17.4%), **8** are currently pending review, and **1** were rejected. The 8 pending request(s) require immediate attention from the blood bank officer to ensure timely patient care.

Internal Requests by Urgency Level

Requests are classified by urgency: **Routine** (non-urgent, planned transfusions), **Urgent** (needed within hours), and **Emergency** (immediate life-saving need). The chart below shows how requests are distributed across these urgency levels, which helps staff understand the acuity of demand and allocate resources accordingly.

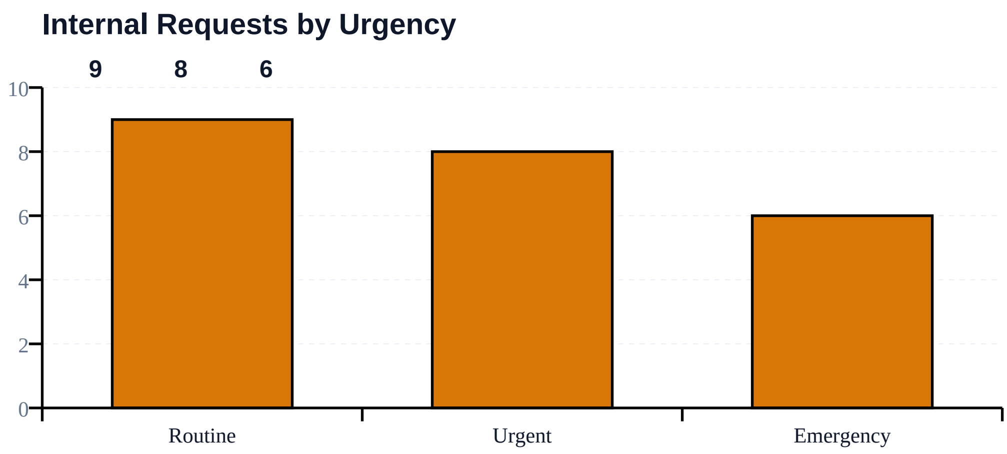


Figure 4: Internal requests by urgency level

Interpretation: 6 emergency request(s) were recorded, indicating critical patient situations that required immediate blood availability. The system's ability to process these rapidly is essential for patient survival outcomes.

4. Network Blood Requests (Broadcast)

The network broadcast feature is the flagship capability of the BBMS platform. When a facility lacks a required blood type, a Blood Bank Officer broadcasts a request to all registered facilities on the network. Facilities with matching available stock submit responses, and the requesting facility selects the most suitable response, reserves the unit, and confirms receipt upon patient arrival. This workflow directly addresses the "blind referral" problem identified in primary research, where up to 57.1% of surveyed facilities could not rule out a failed referral having occurred.

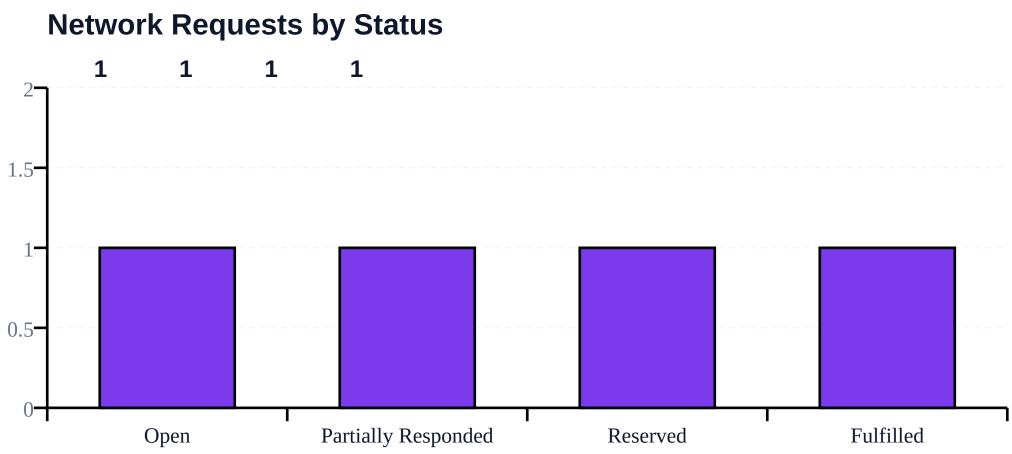


Figure 5: Network broadcast requests by status

Interpretation: 4 network broadcast request(s) were initiated during the reporting period. Of these, **1** were successfully fulfilled (success rate: 25.0%), and **2** are currently open awaiting responses. This demonstrates the system's ability to connect facilities and facilitate cross-facility blood sharing, reducing the risk of failed referrals and preventable patient deaths.

Network Requests by Blood Group

The chart below shows which blood groups are most frequently requested across the network. This information helps facility administrators anticipate demand and prioritize donor recruitment for high-demand groups.

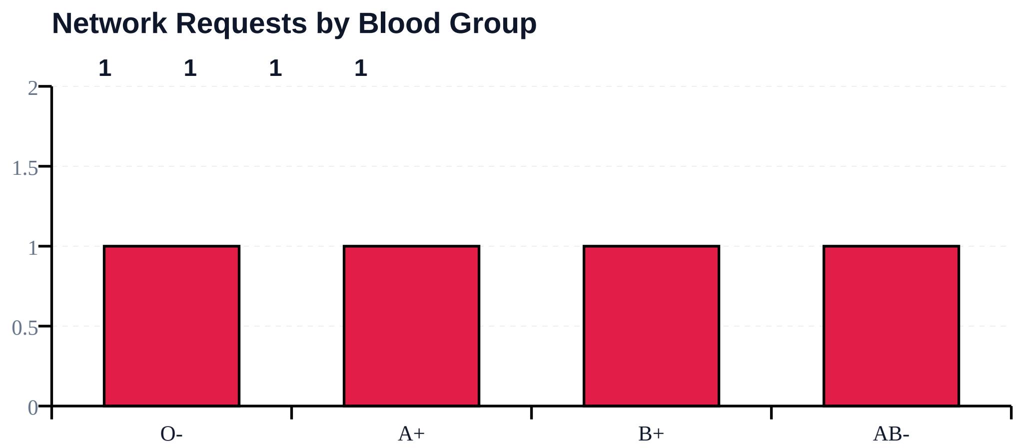


Figure 6: Network requests by blood group

5. Storage Unit Utilization

Proper storage is critical for blood safety. Each storage unit is categorized by temperature: **Refrigerated** (1-6°C for Whole Blood and Red Blood Cells), **Frozen** (-18°C and below for Fresh Frozen Plasma), and **Room Temperature** (20-24°C). The chart below compares current usage against maximum capacity for each storage unit, helping staff identify units that are nearing capacity and may require load redistribution.

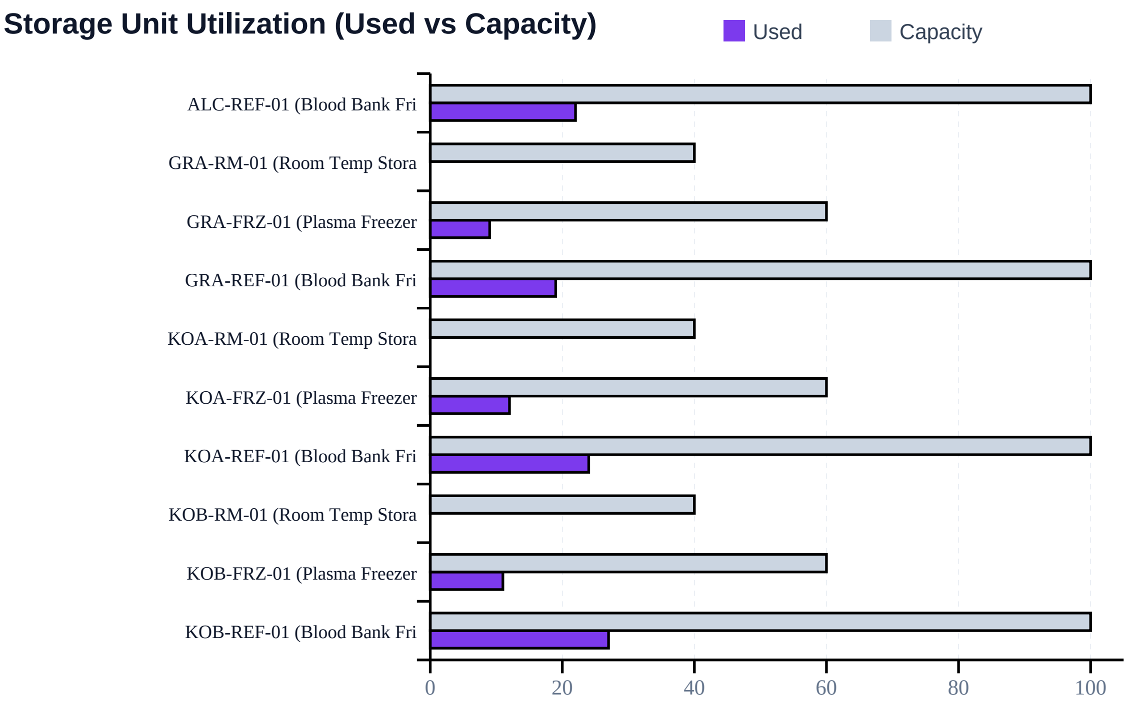


Figure 7: Storage unit utilization (used vs capacity) — page 1 of 2

Storage Unit Utilization (continued, 11-18)

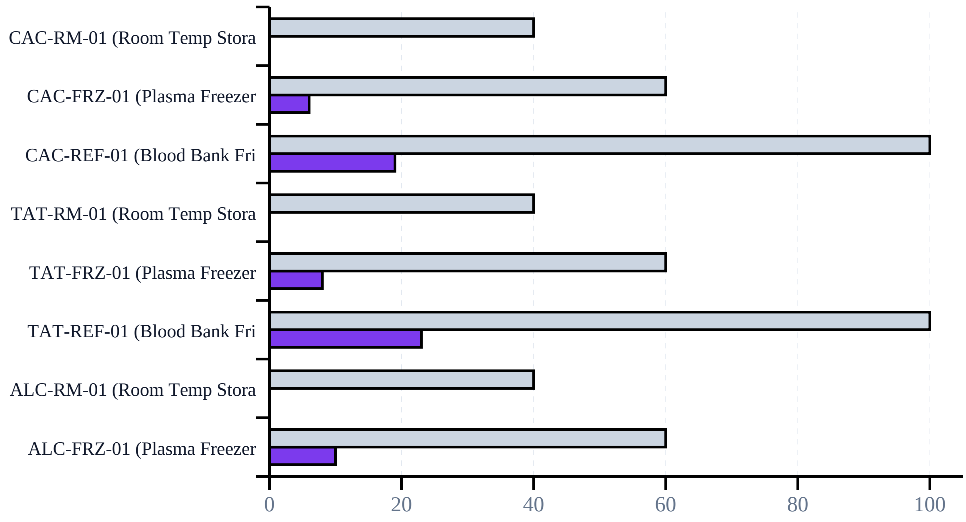


Figure 7b: Storage unit utilization (continued, page 2 of 2)

Detailed Storage Utilization Breakdown

The table below provides exact figures for each storage unit, including the current number of blood units stored, maximum capacity, and utilization percentage. Utilization is color-coded: green for healthy (below 70%), amber for monitoring (70-90%), and red for critical (above 90%).

Storage Unit	Category	Used	Capacity	Utilization
KOB-REF-01 (Blood Bank Fridge)	Refrigerated	27	100	**27%**
KOB-FRZ-01 (Plasma Freezer)	Frozen	11	60	**18%**
KOB-RM-01 (Room Temp Storage)	Room Temperature	0	40	**0%**
KOA-REF-01 (Blood Bank Fridge)	Refrigerated	24	100	**24%**
KOA-FRZ-01 (Plasma Freezer)	Frozen	12	60	**20%**
KOA-RM-01 (Room Temp Storage)	Room Temperature	0	40	**0%**
GRA-REF-01 (Blood Bank Fridge)	Refrigerated	19	100	**19%**
GRA-FRZ-01 (Plasma Freezer)	Frozen	9	60	**15%**
GRA-RM-01 (Room Temp Storage)	Room Temperature	0	40	**0%**
ALC-REF-01 (Blood Bank Fridge)	Refrigerated	22	100	**22%**
ALC-FRZ-01 (Plasma Freezer)	Frozen	10	60	**17%**
ALC-RM-01 (Room Temp Storage)	Room Temperature	0	40	**0%**
TAT-REF-01 (Blood Bank Fridge)	Refrigerated	23	100	**23%**
TAT-FRZ-01 (Plasma Freezer)	Frozen	8	60	**13%**
TAT-RM-01 (Room Temp Storage)	Room Temperature	0	40	**0%**
CAC-REF-01 (Blood Bank Fridge)	Refrigerated	19	100	**19%**
CAC-FRZ-01 (Plasma Freezer)	Frozen	6	60	**10%**
CAC-RM-01 (Room Temp Storage)	Room Temperature	0	40	**0%**

Table 2: Detailed storage utilization breakdown

Interpretation: All storage units are operating below 70% capacity, indicating healthy storage headroom for incoming donations.

6. Recent Activity (Audit Trail)

The BBMS platform maintains a tamper-resistant audit log of all significant system actions, including user logins, blood unit registrations, status changes, request processing, and network request activity. Audit logs are write-once and cannot be modified or deleted by any user role, ensuring a complete and verifiable record for regulatory compliance under Ghana's Data Protection Act 2012 (Act 843). The table below shows the most recent activity entries.

Timestamp	Action	Description	User
2026-07-08 11:06	LOGIN	System Administrator logged in	System Administrator
2026-07-08 11:04	LOGIN	Hospital Administrator (Korle Bu Teaching Hospital) log	Hospital Administrator
2026-07-08 10:57	LOGIN	System Administrator logged in	System Administrator
2026-07-08 10:42	LOGIN	Hospital Administrator (Korle Bu Teaching Hospital) log	Hospital Administrator
2026-07-08 10:36	LOGIN	Hospital Administrator (Korle Bu Teaching Hospital) log	Hospital Administrator
2026-07-08 09:49	LOGIN	System Administrator logged in	System Administrator
2026-07-07 22:13	LOGOUT	System Administrator logged out	System Administrator
2026-07-07 22:12	LOGIN	System Administrator logged in	System Administrator
2026-07-07 22:05	LOGIN	Blood Bank Officer (Korle Bu Teaching Hospital) logged	Blood Bank Officer (Ko
2026-07-07 22:05	LOGOUT	System Administrator logged out	System Administrator
2026-07-07 22:03	LOGIN	System Administrator logged in	System Administrator
2026-07-07 22:03	LOGOUT	Blood Bank Officer (Korle Bu Teaching Hospital) logged	Blood Bank Officer (Ko

Table 3: Recent system activity (most recent 12 entries)

Report Generated

This report was generated by the BBMS Ghana Cloud-Based Blood Bank Management System on 08 July 2026 at 11:11 for All Registered Facilities. The data presented reflects system records as of the generation timestamp. For questions or corrections, contact the System Administrator.

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